

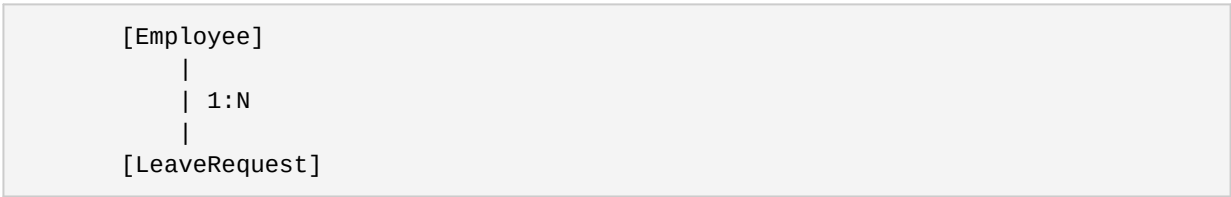
Data Model Documentation

Overview

The system manages the following core domains:

- Employee: Responsible for storing employee details and their leave requests.
- LeaveRequest: Responsible for tracking the details of each leave request submitted by employees.

Entity Relationship Diagram



Entities

Employee

Purpose: Represents an employee within the organization, including their personal details and leave balance.

Table/Collection Name: Employee

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier for the employee
name	VARCHAR	Yes	Max 255 chars	Employee's full name
email	VARCHAR	Yes	Max 255 chars, unique	Employee's email address
leave_balance	INTEGER	Yes	Min 0	Employee's remaining leave balance

Relationships:

- Has many LeaveRequest (1:N): Inverse via LeaveRequest . employee_id

Validation Rules:

- email must be non-empty and unique across all records
- leave_balance must be a non-negative integer

LeaveRequest

Purpose: Represents a leave application submitted by an employee, including details such as the leave type, start and end dates, and status.

Table/Collection Name: LeaveRequest

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier for the leave request
employee_id	UUID	Yes	Foreign key to Employee . id	Employee who submitted the request
leave_type	VARCHAR	Yes	Must be one of "SICK", "VACATION", "PERSONAL"	Type of leave requested
start_date	DATE	Yes	Must be a valid date	Start date of the leave
end_date	DATE	Yes	Must be a valid date and after start_date	End date of the leave
status	VARCHAR	Yes	Must be one of "PENDING", "APPROVED", "REJECTED"	Current status of the leave request
created_at	TIMESTAMP	Yes	Auto-set on create	Timestamp when the request was created
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to Employee (N:1): Foreign key employee_id references Employee.id

Validation Rules:

- leave_type must be one of "SICK", "VACATION", "PERSONAL"
- start_date and end_date must be valid dates with end_date after start_date

Relationships and Constraints

Foreign Keys

From Entity	Field	To Entity	Field	Cardinality	Delete Behavior
LeaveRequest	employee_id	Employee	id	N:1	CASCADE

Unique Constraints

- Employee(email): Ensures no duplicate email addresses

Data Validation Rules

Business Rules:

- leave_balance: Must be a non-negative integer
- start_date and end_date: end_date must be after start_date

Format Rules:

- email: Must match the regex `^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$`
- leave_type: Must be one of "SICK", "VACATION", "PERSONAL"

Data Lifecycle

Creation

- A new Employee is created when a new user is onboarded
- Default leave_balance is set to 12 (assuming 12 days of leave per year)
- created_by is automatically set to the HR manager

Updates

- Employee.leave_balance can only be updated by the HR department
- updated_at is automatically updated on any modification

Archival / Soft Delete

- Records are not archived but can be marked as inactive by setting status to "INACTIVE"

Indexes

- Primary key: id
- Foreign keys: employee_id
- Search fields: name, email

Example Records

Employee

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "name": "John Doe",
  "email": "john.doe@example.com",
  "leave_balance": 12
}
```

LeaveRequest

```
{
  "id": "123e4567-e89b-12d3-a456-426614174000",
  "employee_id": "550e8400-e29b-41d4-a716-446655440000",
  "leave_type": "SICK",
  "start_date": "2023-10-01",
  "end_date": "2023-10-05",
  "status": "PENDING",
  "created_at": "2023-09-30T10:30:00Z",
  "updated_at": "2023-09-30T10:30:00Z"
}
```